The Central Distribution: A Minimax-KL Framework for Representing Partial Knowledge

Behrouz Ayati
aybehrouz@gmail.com

July 11, 2025

Abstract

We introduce the *central distribution*, a probability distribution that represents a hierarchy of probability distributions by minimizing the worst-case Kullback–Leibler (KL) divergence to the admissible models. This provides a principled method for converting partial knowledge and set-valued uncertainty into a single probability measure suitable for Bayesian inference.

We show that central distributions admit a minimax characterization and can be obtained as solutions to convex–concave saddle point problems. For parametric models, we establish a sufficient fixed-point condition for priors whose induced marginal distributions are central, establishing a formal connection with Bernardo's reference prior methodology. We extend the framework to hierarchical uncertainty structures and propose an asymptotic notion of centrality for infinite sequences.

Finally, we illustrate how uncertainty about relevant feature subsets can be represented hierarchically, and how central distributions can be used to aggregate such structural uncertainty into a single predictive model. This suggests an information-theoretic framework for constructing noninformative priors and performing Bayesian feature selection.

1 Introduction

We begin by revisiting a fundamental decision problem in Bayesian decision theory. An agent must select an action from a set of available actions. The resulting loss depends on an unknown state of the environment and is quantified by a real-valued loss function. Specifically, let a denote an action and x the unknown state. The loss associated with choosing action a when the state is x is given by the loss function $\ell(a, x)$.

We assume that some evidence e about the unknown state is available and can be used to guide the agent's decision. In other words, the agent aims to find an optimal strategy, where a strategy is a conditional distribution over actions given the observed evidence, denoted by $S(A \mid E)$.

In Bayesian decision theory, the optimal strategy is the one that minimizes the expected loss:

$$S_{\text{opt}} = \arg\min_S \sum_a \sum_{x,e} \ell(a, x) S(a \mid e) P(x, e).$$

One can show that there always exists a deterministic optimal strategy, namely a strategy that, for each evidence e, selects the action minimizing the expected loss conditioned on e:

$$S_{\text{opt}}(e) = \arg\min_a \mathbb{E}_P \left[\ell(a, x) \mid e \right].$$

Now suppose that, instead of basing our decisions on the original evidence E, we choose to use a transformed version E'. In practice, the evidence available to the agent is fixed by the environment. However, the agent may choose to base its decisions on a deterministic transformation $E' = f(E)$. This is analogous to feature selection in machine learning, where a high-dimensional observation is replaced by a lower-dimensional representation.

Could such a transformation improve the agent's strategy? Within the framework of Bayesian decision theory, this question has a straightforward answer. Any two strategies can be compared through their expected loss: the strategy with the lower expected loss is preferred. However, a fundamental result of Bayesian decision theory states that the optimal strategy based on E is always at least as good as any strategy based on E'.

One might argue that adopting a strategy that uses simpler evidence may discard information and is therefore not justified by Bayesian decision theory. The key point here is that the space of strategies using E' is a subset of the space of strategies using E. When we search for the optimal strategy based on E, we are simultaneously considering all strategies that depend on E', since every such strategy can also be expressed as a strategy based on E. This helps explain why standard Bayesian inference does not inherently favor simpler representations of the evidence.

With an appropriate choice of prior distribution, the Bayesian framework may effectively favor strategies that depend only on relevant aspects of the evidence. In this sense, the prior can induce a form of feature selection. However, many commonly used noninformative priors, such as flat priors, do not appear to naturally promote feature selection.

In this paper, we introduce the central distribution, a minimax-KL representative of a set, or more generally a hierarchy, of probability distributions. This provides an information-theoretic framework for converting set-valued and hierarchical uncertainty into a single probability distribution suitable for Bayesian inference. We establish existence, uniqueness, and optimization-theoretic characterizations of central distributions, and develop the corresponding notion of central priors for parametric models. We further identify a sufficient fixed-point condition linking central priors to objective Bayesian reference analysis [3]. Finally, we investigate asymptotic extensions of the framework and illustrate how hierarchical central distributions can be used to represent uncertainty about model structure in a feature-selection setting.

1.1 Related Work

The problem of constructing objective or noninformative priors has a long history in Bayesian statistics. Among the most influential approaches are Jeffreys priors, reference priors, and maximum-entropy based methods [6, 2, 5].

Jeffreys introduced a general rule for constructing invariant noninformative priors based on the Fisher information matrix [6]. The resulting priors are widely used because of their parameterization invariance and their connection to asymptotic likelihood theory. However, Jeffreys priors are derived from local geometric properties of the parameter space and do not directly address the problem of representing a set of admissible probability distributions by a single distribution.

The closest connection to the present work is Bernardo's reference prior methodology [3, 2]. Reference priors are constructed by maximizing the expected information gained from the data and are intended to represent a state of minimal prior information. In Section 2.3, we show that central priors satisfy a fixed-point condition closely related to the reference prior construction. In particular, for a single parameter and a single set of distributions, the resulting central prior coincides with the corresponding reference prior.

Several authors have also studied objective Bayesian inference through information-theoretic principles [5, 8]. Examples include maximum entropy methods, minimum discrimination information principles, and related divergence-based approaches. These methods typically select a distribution by optimizing an information criterion subject to constraints. In contrast, the central distribution is defined as a minimax representative of a set or hierarchy of distributions, minimizing the worst-case Kullback–Leibler divergence to the admissible models.

The present work is also related to robust Bayesian statistics and imprecise probability [9, 10]. In these frameworks uncertainty is represented by sets of probability distributions rather than a single model. Existing approaches generally focus on performing inference directly with sets of distributions. The central distribution framework instead seeks a single representative distribution that summarizes the available set-valued or hierarchical uncertainty while preserving its information-theoretic structure.

The central distribution framework differs from these approaches in two respects. First, it is defined as a minimax-KL representative of a set, or more generally a hierarchy, of probability distributions. Second, it provides a mechanism for converting set-valued and hierarchical uncertainty into a single probability distribution suitable for Bayesian inference. While related to objective Bayesian inference, robust Bayesian analysis, and information-theoretic methods, its primary goal is not the direct specification of a prior distribution for parameters or inference with sets of distributions, but rather the construction of a representative distribution that preserves the informational structure of the underlying uncertainty.

1.2 Divergence

Suppose that for a given environment, we have specified a probability measure P. In particular, we are interested in the distribution it induces on a random variable X, denoted $P(X)$.

Now, imagine we ask an expert—a person familiar with the environment—to evaluate our choice of P. The expert may hold different beliefs and instead favor another probability measure Q. From the expert's perspective, evaluating P involves assessing the potential loss incurred by using the (in their view) incorrect distribution $P(X)$ when the true distribution is believed to be $Q(X)$.

Let us assume that a real number can be used to measure this loss, and denote it as $\mathbb{D}\left[Q(X) \parallel P(X)\right]$. It seems reasonable, as P gets closer to Q this measure decreases, and increases when P diverges from Q. Therefore, $\mathbb{D}$ effectively is a measure of the divergence of $P(X)$ from $Q(X)$, when Q is the believed true probability measure.

Interestingly, if we make certain assumptions about the properties of this divergence measure, it turns out that—up to a constant factor—it must be equal to the Kullback—Leibler (KL) divergence.

Alternatively, by adopting a more minimal set of properties, one can prove that accepting Shannon's entropy as a measure of the uncertainty implies that KL divergence can be considered as an appropriate method for measuring divergence between distributions.

Theorem 1. *Let X be a discrete random variable. Suppose $\mathbb{D}\left[Q(X) \parallel P(X)\right]$ is some divergence measure that quantifies the divergence of $P(X)$ from $Q(X)$. Consider the following properties:*

Consistency with entropy *Let $\mathit{\Omega}$ be a continuous random variable, and Q be a probability measure. If $U(\mathit{\Omega})$ is the uniform distribution with the same support as $Q(\mathit{\Omega})$, then we have*

$$\mathbb{D}\left[Q(\mathit{\Omega}) \parallel U(\mathit{\Omega})\right] = \mathbb{H}\left[U(\mathit{\Omega})\right] - \mathbb{H}\left[Q(\mathit{\Omega})\right],$$

where $\mathbb{H}$ denotes Shannon's entropy.

Additivity *For any two discrete random variables X, Y and probability measures P, Q*

$$\mathbb{D}\left[Q(X,Y) \parallel P(X,Y)\right] = \mathbb{D}\left[Q(X) \parallel P(X)\right] + \sum_x \mathbb{D}\left[Q(Y|x) \parallel P(Y|x)\right] Q(x)\cdot$$

If $\mathbb{D}\left[\,\cdot\,\right]$ satisfies these properties, then it is the Kullback–Leibler divergence.

Proof. Let $P(X)$ and $Q(X)$ be two arbitrary distributions for a discrete random variable X. We define a new continuous random variable $\mathit{\Omega}$ and let

$$p(\mathit{\Omega} = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \leq \omega \leq P(X = x), \\ 0 & \text{otherwise}\cdot \end{cases}$$

Notice that both $P(\mathit{\Omega}|X)$ and $P(\mathit{\Omega}, X)$ are uniform distributions, and

$$\mathbb{H}\left[P(\mathit{\Omega}|x)\right] = \ln P(x),\ \mathbb{H}\left[P(\mathit{\Omega}, X)\right] = 0\cdot$$

We define $Q(\mathit{\Omega}|x)$ to be an arbitrary distribution that has the same support as $P(\mathit{\Omega}|x)$. This ensures that $Q(\mathit{\Omega}, X)$ and $P(\mathit{\Omega}, X)$ have the same support as well. Thus, we have

$$\begin{aligned} \mathbb{D}\left[Q(\mathit{\Omega}, X) \parallel P(\mathit{\Omega}, X)\right] &= \mathbb{H}\left[P(\mathit{\Omega}, X)\right] - \mathbb{H}\left[Q(\mathit{\Omega}, X)\right] \\ &= \sum_x \int q(\omega, x) \ln q(\omega, x) d\omega \\ &= \sum_x Q(x) \int q(\omega \mid x) \ln q(\omega, x) d\omega, \end{aligned}$$

and

$$\begin{aligned}\mathbb{D}\left[Q(\Omega|x) \parallel P(\Omega|x)\right] &= \mathbb{H}\left[P(\Omega|x)\right] - \mathbb{H}\left[Q(\Omega|x)\right] \\ &= \ln P(x) + \int q(\omega \mid x) \ln q(\omega \mid x) d\omega\cdot\end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$\begin{aligned}\mathbb{D}\left[Q(X) \parallel P(X)\right] &= \mathbb{D}\left[Q(\Omega, X) \parallel P(\Omega, X)\right] - \sum_x \mathbb{D}\left[Q(\Omega|x) \parallel P(\Omega|x)\right] Q(x) \\ &= \sum_x Q(x) \left(\int q(\omega \mid x) \ln \frac{q(\omega, x)}{q(\omega \mid x)} d\omega - \ln P(x) \right) \\ &= \sum_x Q(x) \left(\ln Q(x) \int q(\omega \mid x) d\omega - \ln P(x) \right) \\ &= \sum_x Q(x) \ln \frac{Q(x)}{P(x)}\cdot\end{aligned}$$

□

In this theorem, the first property simply states that the entropy of a distribution should be linearly related to the divergence of the uniform distribution from it. The more the divergence is, the less the entropy must be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(X, Y)$ from $Q(X, Y)$ is a measure of the generic loss of using $P(X, Y)$ when we believe $Q(X, Y)$ is the true distribution. To calculate this loss, we can assume that X happens before Y. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with the probability $Q(x)$, it would also include the loss of using $P(Y \mid X = x)$. This property indicates that for combining these individual terms we should use addition.

We now briefly review some KL divergence properties needed in subsequent sections.

Lemma 1. *Let X be a random variable, and let P, Q, and R be probability distributions over X. The Kullback—Leibler divergence satisfies the following inequality:*

$$\mathbb{D}_{\text{KL}}\left[Q(X) \parallel P(X)\right] \geq \sum_x Q(x) \ln \frac{R(x)}{P(x)}\cdot$$

Proof. The Kullback—Leibler divergence is nonnegative and is defined when R is a probability distribution:

$$\mathbb{D}_{\text{KL}}\left[Q(X) \parallel R(X)\right] \geq 0\cdot$$

By adding $\sum_x Q(x) \ln \frac{R(x)}{P(x)}$ to both sides of the above inequality, the desired inequality follows. $\square$

Theorem 2. *Let Q_1, Q_2 and P be probability distributions. If for some $0 \leq \alpha \leq 1$ we have $Q = \alpha Q_1 + (1-\alpha)Q_2$, then:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q \,\|\, P\right] \leq \alpha \mathbb{D}_{\mathrm{KL}}\left[Q_1 \,\|\, P\right] + (1-\alpha)\mathbb{D}_{\mathrm{KL}}\left[Q_2 \,\|\, P\right] \cdot$$

That is, the KL divergence is convex in its first argument.

Proof. The result follows from Lemma 1 after expanding the left-hand side of the inequality using the definition of KL divergence. $\square$

The KL divergence plays a key role in the notion of central distribution, which we introduce in the next section.

2 Central Distribution

Consider a random variable X. In Bayesian modeling, our knowledge about X is represented by a probability distribution, which serves as a basis for making inferences.

However, in many situations, our information may be too limited to justify choosing a single distribution. Instead, we may be able to specify a set of plausible distributions. Such a set represents some form of knowledge: although it does not identify a single distribution, it constrains the range of plausible probabilistic models.

A well-known example of this idea is the exchangeability assumption used in de Finetti's theorem. A sequence of random variables $X_1, X_2, \ldots$ is said to be exchangeable if the ordering of the variables carries no information. That is, the joint probability distribution remains invariant under permutations of the variables. This assumption imposes a constraint, effectively introducing a set of plausible distributions. Thus, exchangeability can be seen as a specific form of partial knowledge.

In this work, we adopt the view that a set of probability distributions can represent a meaningful state of *uncertainty* or *partial knowledge*, and that such set-valued uncertainty may itself be used as the basis for inference.

More generally, this partial knowledge may possess a hierarchical structure. In the simplest case, uncertainty is represented by a single set of candidate distributions. In richer settings, however, this uncertainty may be organized into nested collections, where elements of a set may themselves be sets of distributions. Such nesting can continue recursively, yielding a hierarchical structure.

Crucially, the choice of hierarchical organization is itself informative: grouping distributions into subsets expresses that certain alternatives are regarded as belonging to a common epistemic category and should therefore be aggregated before being compared with alternatives in other groups. Therefore, the hierarchy encodes information beyond the collection of leaf distributions alone.

This organization may be interpreted as introducing multiple layers of latent structural decisions, where higher-level choices restrict the admissible lower-level alternatives.

Within a Bayesian framework, inference ultimately requires a single probability measure. Hence, to use hierarchical partial knowledge in Bayesian inference, we require a principled method for transforming a nested set-valued uncertainty into a single probability distribution while preserving the informational structure encoded by the hierarchy.

From a minimax perspective, when uncertainty is represented by a set of candidate distributions, a natural representative distribution is one that minimizes the worst-case divergence to the members of the set. Such a distribution may be viewed as the most conservative or robust single-distribution summary of the available knowledge.

Notation 2.1 (Kullback–Leibler Divergence)**.** For probability distributions P and Q, $\mathbb{D}_{\mathrm{KL}}[Q(X) \,\|\, P(X)]$ denotes the Kullback–Leibler divergence between P and Q over a random variable X. For a discrete random variable X we have

$$\mathbb{D}_{\mathrm{KL}}[Q(X) \,\|\, P(X)] = \sum_{x} Q(x) \ln \frac{Q(x)}{P(x)}.$$

When the underlying random variable is clear from context, we omit the argument and write $\mathbb{D}_{\mathrm{KL}}[Q \,\|\, P]$.

We assume that the KL divergence is defined for every pair of probability distributions P and Q, taking the value $+\infty$ whenever the support of Q is not contained in the support of P.

Definition 2.1. Let X be a random variable and $\mathcal{Q}$ be a set of probability distributions over X. We call a distribution C a *central distribution* of $\mathcal{Q}$ for X if, for every probability distribution P, we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}[Q(X) \,\|\, C(X)] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}[Q(X) \,\|\, P(X)], \tag{1}$$

By convention, any probability distribution is regarded as its own central distribution.

Under this definition, the central distribution of $\mathcal{Q}$ is not necessarily an element of $\mathcal{Q}$. That is, the distribution that best represents the set in the specified sense may lie outside the set it summarizes.

Sometimes partial knowledge is represented by a hierarchical structure of sets of distributions. Our current definition of the central distribution does not account for this case. The following recursive definitions extend the concept to handle such situations. While not directly about feature selection, this extension plays a key role in enabling the connection between central distributions and feature selection, a connection that will become clearer later.

Definition 2.2 (Hierarchy of Distributions)**.** A *hierarchy of distributions* over X is defined recursively as follows:

- Any probability distribution over X is a hierarchy of distributions over X.
- Any set whose elements are hierarchies of distributions over X is itself a hierarchy of distributions over X.

Intuitively, a central distribution acts as a representative or summary of a set of distributions. While it may not belong to the set itself, it reflects the overall properties of the set in a way that is meaningful for inference. It also provides a way to evaluate probability distributions under partial knowledge, motivating the definition of a corresponding central divergence.

The concept of a central distribution is not intended to yield a unique object; rather, all distributions satisfying the definition can be regarded as equally good representatives. However, when multiple central distributions exist, our constructions should not depend on an arbitrary choice among them. To avoid such dependence, we define the central divergence using the best divergence attained over the set of all central distributions.

Definition 2.3 (Central Divergence)**.** For any distribution P, we define the *central divergence* of P from a hierarchy of distributions $\mathcal{H}$, with respect to a random variable X, as follows:

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{H}(X) \,\|\, P(X)\right] = \inf_{C} \mathbb{D}_{\mathrm{KL}}\left[C(X) \,\|\, P(X)\right],$$

where C denotes a central distribution of $\mathcal{H}$ for X.

So far, this definition applies only when the hierarchy is a single set of distributions. The following definition extends the construction recursively to general hierarchical uncertainty structures.

Definition 2.4 (Central Distribution)**.** Let X be a random variable, and let $\mathcal{H}^*$ be a hierarchy of distributions over X. We call a distribution C a *central distribution* of $\mathcal{H}^*$ if, for every probability distribution P, we have:

$$\sup_{\mathcal{H}\in\mathcal{H}^*} \mathbb{D}_{\mathrm{C}}\left[\mathcal{H}(X) \,\|\, C(X)\right] \leq \sup_{\mathcal{H}\in\mathcal{H}^*} \mathbb{D}_{\mathrm{C}}\left[\mathcal{H}(X) \,\|\, P(X)\right].$$

Definition 2.4 generalizes Definition 2.1, which is recovered when the hierarchy is a set of probability distributions.

For later use, we introduce a compact notation for the minimax objective appearing in Definition 2.1.

Notation 2.2. For a set of probability distributions $\mathcal{Q}$ over X, we define the functional

$$\mathbb{S}_{\mathcal{Q}}\left[P(X)\right] \coloneqq \sup_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right].$$

When $\mathcal{Q}$ and X are clear from the context, we may simply write $\mathbb{S}\left[P\right]$.

This notation allows Definition 2.1 to be written compactly as

$$C \in \arg\min_{P} \mathbb{S}_{\mathcal{Q}}\left[P(X)\right].$$

A simple example illustrates that hierarchical organization can affect the resulting representative distribution even when the underlying admissible sets remain unchanged.

Example 2.1. Let X be a random variable with three possible outcomes x_1, x_2, x_3. For each i, let S_i denote the set of all probability distributions over X that assign zero probability to x_i.

By symmetry, the central distribution of each set S_i is the uniform distribution over its support:

$$C_{S_1} = (0, 0.5, 0.5), \qquad C_{S_2} = (0.5, 0, 0.5), \qquad C_{S_3} = (0.5, 0.5, 0).$$

If the three sets S_1, S_2, S_3 are treated symmetrically as elements of a single set, then their central distribution is

$$C_{\{S_1,S_2,S_3\}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

Now consider instead a hierarchical organization in which S_1 and S_2 are grouped together, while S_3 forms its own separate branch.

By numerically solving the minimax problem in Definition 2.4, we obtain

$$C_{\{S_1,S_2\}} = (0.25, 0.25, 0.5).$$

Applying the same procedure to the two distributions $C_{\{S_1,S_2\}}$ and C_{S_3} yields

$$C_{\{\{S_1,S_2\},S_3\}} = (0.4, 0.4, 0.2)\,.$$

Although the underlying admissible sets are identical in both constructions, the resulting representative distribution differs.

This demonstrates that hierarchical structure carries information beyond the collection of leaf distributions alone. Consequently, flattening a hierarchy into a single unstructured set may discard meaningful epistemic structure.

In many learning problems, uncertainty concerns not only the parameters of a predictive model but also its structural form. In particular, when the relevance of potential features is unknown, each candidate feature subset determines a distinct set of plausible models in which the target variable depends only on that feature subset.

Feature selection may therefore be viewed as uncertainty over a collection of distributions. The hierarchical central distribution framework provides a principled mechanism for representing and aggregating such uncertainty. By organizing candidate models hierarchically, one can encode preferences over structural hypotheses, such as favoring sparse subsets, grouping related attributes, or privileging particular interactions.

Candidate feature subsets typically induce overlapping families of admissible distributions. Distributions representing simpler dependency structure may belong simultaneously to many such families, whereas more complex dependency patterns belong to fewer. As a result, hierarchical aggregation through central distributions can induce a preference toward structurally simpler models by naturally assigning more weight to distributions compatible with multiple structural hypotheses.

2.1 Minimax Characterization

With the concept of hierarchical central distributions in place, we now develop the theoretical machinery needed to characterize and compute them. Throughout this subsection, X denotes a discrete random variable with finite support and $\mathcal{Q}$ denotes a finite set of probability distributions over X.

As we shall see, central distributions arise naturally as solutions to a convex–concave optimization problem. By formulating the appropriate objective function, we obtain a saddle-point characterization that enables the use of standard techniques from convex optimization.

Definition 2.5 (Saddle Point)**.** Consider a function $f : \mathcal{W} \times \mathcal{Z} \to \mathbb{R}$. We refer to a pair (w^*, z^*) as a *saddle point* for f if we have

$$f(w^*, z) \leq f(w^*, z^*) \leq f(w, z^*),$$

for all $w \in \mathcal{W}$ and $z \in \mathcal{Z}$.

Lemma 2. *Define:*

$$f(P, R) = \sum_i \mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \parallel P(X)\right] R(i),$$

where P and R are probability distributions, $Q_i \in \mathcal{Q}$ and the index i ranges over all elements of $\mathcal{Q}$. Then there exists a saddle point for f.

Proof. The Kullback–Leibler divergence is convex and lower-semicontinuous in its second argument. Therefore, for any fixed probability distribution R, the function $f(\,\cdot\,, R)$ is a finite convex combination of lower-semicontinuous convex functions and is therefore itself convex and lower-semicontinuous in P.

For fixed P, the function $f(P, \,\cdot\,)$ is affine in R, and hence concave and continuous.

The set of probability distributions over a discrete random variable with finite support is a compact convex simplex. Therefore, the assumptions of Sion's minimax theorem are satisfied [7], and

$$\min_P \sup_R f(P, R) = \sup_R \min_P f(P, R).$$

Since $f(P, \,\cdot\,)$ is continuous and the feasible set for R is compact, the supremum is attained for every P. Hence the minimax solution exists and f admits a saddle point. $\square$

Lemma 3. *Let f be the function defined in Lemma 2. If (P^*, R^*) is a saddle point for f, then P^* is a central distribution of $\mathcal{Q}$ for X.*

Proof. Let P be an arbitrary probability distribution over X, and (P^*, R^*) be a saddle point for f. As P is a probability distribution, $f(P, R^*)$ is defined, and by the definition of a saddle point

$$f(P^*, R^*) \leq f(P, R^*). \tag{2}$$

Since R^* is a probability distribution, the definition of f implies:

$$f(P, R^*) \leq \mathbb{S}\left[P\right].$$

On the other hand, by the definition of a saddle point:

$$f(P^*, R^*) = \max_R f(P^*, R) = \mathbb{S}\left[P^*\right].$$

Hence, Equation 2 implies

$$\mathbb{S}\left[P^*\right] \leq \mathbb{S}\left[P\right].$$

This holds for any arbitrary P. Therefore, by Definition 2.1, P^* is a central distribution of $\mathcal{Q}$ for X. $\square$

Lemma 4. *Let f be the function defined in Lemma 2. The point (P^*, R^*) is a saddle point for f, if and only if there is a constant λ, and a function μ, such that for every x and i the following conditions hold:*

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} + \mu(i) = \lambda, \tag{3}$$

$$P^*(x) = \sum_i R^*(i) Q_i(x) \tag{4}$$

$$\sum_i R^*(i) = 1, \tag{5}$$

$$R^*(i) \geq 0, \tag{6}$$

$$\mu(i) \geq 0, \tag{7}$$

$$\mu(i) R^*(i) = 0. \tag{8}$$

Furthermore, for a fixed probability distribution R, a distribution $\hat{P}$ minimizes $f(\,\cdot\,, R)$, if and only if for all x it satisfies

$$\hat{P}(x) = \sum_i Q_i(x) R(i). \tag{9}$$

Proof. We extend f to a function $\hat{f}$ defined as follows:

$$\hat{f}(P, R) = \sum_i R(i) \sum_x Q_i(x) \ln \frac{Q_i(x)}{P(x)},$$

where the domain is extended by relaxing the constraints that P and R be probability distributions. This enables us to formulate the saddle points of f as solutions to a constrained optimization problem defined for $\hat{f}$.

Let (P^*, R^*) be a simultaneous solution to the following two optimization problems:

$$
\begin{array}{lll}
\text{minimize} & \hat{f}(P,R) & \text{w.r.t. } P \\
\text{subject to} & \sum_x P(x) = 1, & \\
& & \\
\text{maximize} & \hat{f}(P,R) & \text{w.r.t. } R \\
\text{subject to} & R(i) \geq 0, & \text{for all } i \\
& \sum_i R(i) = 1 &
\end{array}
\tag{10}
$$

If (P^*, R^*) exists, it will be a saddle point for f.

Both optimization problems are convex and differentiable; see [4]. To establish this, consider that all the constraint functions are affine. Additionally, both $\hat{f}(P,\,\cdot\,)$ and $\hat{f}(\,\cdot\,,R)$ are differentiable on their domains. For any fixed P, $\hat{f}(P,\,\cdot\,)$ is an affine function and is concave. For a fixed solution to the second optimization problem denoted by $\hat{R}$, we can write

$$
\hat{f}(P,\hat{R}) = c - \sum_{i,x} \hat{R}(i) Q_i(x) \ln P(x), \tag{11}
$$

where c is not a function of P. Since $\hat{R}$ is a solution to the second optimization problem, it satisfies the corresponding constraints and is therefore a valid probability distribution. Hence, for all i, x we have $\hat{R}(i)Q_i(x) \geq 0$, and $\hat{f}(\,\cdot\,,\hat{R})$ is the negative of a linear combination of concave $\ln P(x)$ functions with nonnegative coefficients. This function is therefore convex.

Thus, both optimization problems are convex and differentiable, with affine equality and inequality constraints. In this setting, Slater's condition reduces to feasibility [4]. The feasible set for the first problem is the probability simplex, while the feasible set for the second problem is the strictly positive probability simplex. Both sets are nonempty. Therefore, Slater's condition holds, and the KKT conditions are necessary and sufficient for optimality. In other words, if there exists a point (P^*, R^*) that satisfies the KKT conditions for both problems, it will be a saddle point for f, and vice versa.

The KKT conditions are as follows. There exist constants λ, η and a function μ, such that for every x and i:

$$\sum_i R^*(i) Q_i(x) = \eta P^*(x) \tag{12}$$

$$\sum_x P^*(x) = 1, \tag{13}$$

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} + \mu(i) = \lambda,$$

$$\sum_i R^*(i) = 1, \tag{14}$$

$$R^*(i) \geq 0,$$

$$\mu(i) \geq 0,$$

$$\mu(i) R^*(i) = 0.$$

By summing Equation 12 over x and using Equation 14, we obtain $\eta = 1$. Substituting this into Equation 12 yields Equation 4, which in turn implies Equation 13.

To prove the second part of the lemma, we note that Equations 12 and 13 are the KKT conditions for the minimization problem in Equation 10. As shown above, the KKT conditions are necessary and sufficient for optimality, and when R is a probability distribution, these conditions simplify to

$$\hat{P}(x) = \sum_i Q_i(x) R(i). \qquad \square$$

Lemma 5. *Let f be the function defined in Lemma 2. Define the function g as:*

$$g(R) = \sum_i \mathbb{D}_{\mathrm{KL}} \left[Q_i(X) \parallel R(X) \right] R(i),$$

where $Q_i \in \mathcal{Q}$, and R is a probability distribution such that

$$R(X, I) = Q_I(X) R(I).$$

If (P^*, R^*) is a saddle point for f, then the joint distribution $Q_I(X) R^*(I)$ is a maximizer of g.

Proof. Let (P^*, R^*) be a saddle point for f. By Lemma 4, P^* satisfies Equation 4. Letting $R^*(X, I) = Q_I(X) R^*(I)$ we have

$$g(R^*) = f(P^*, R^*).$$

Now, let R be an arbitrary probability distribution such that $g(R)$ is defined. From the second part of Lemma 4, it follows that $g(R)$ indeed corresponds to the minimum of $f(\,\cdot\,, R)$, which implies

$$g(R) \leq f(P^*, R).$$

Since (P^*, R^*) is a saddle point, we also have

$$f(P^*, R) \leq f(P^*, R^*).$$

Combining the two inequalities, we get:

$$g(R) \leq g(R^*).$$

This shows that $g(R^*)$ is the maximum of g, as desired. □

The following theorem establishes the existence and uniqueness of the central distribution and provides a characterization of its optimality in the KL-minimax setting. This characterization motivates the term *central distribution*: like the center of a circle, the central distribution is equidistant from the boundary.

Theorem 3 (Centrality)**.** *Let X be a discrete random variable with finite support, and let $\mathcal{Q}$ be a finite set of probability distributions over X. Then $\mathcal{Q}$ has a unique central distribution for X.*

Moreover, C is the central distribution of $\mathcal{Q}$ for X if and only if it can be expressed as a convex combination of the elements of $\mathcal{Q}$; that is,

$$C(x) = \sum_i Q_i(x) R(i),$$

where R is a probability distribution over the indices of Q, and there exists a constant λ such that

$$\mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \,\|\, C(X)\right] = \lambda$$

whenever $R(i) > 0$, and

$$\mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \,\|\, C(X)\right] \leq \lambda$$

whenever $R(i) = 0$.

Proof. By Lemma 2 the function f has a saddle point. Let (C, R) denote this saddle point. By Lemma 3, C is a central distribution of $\mathcal{Q}$ for X.

The saddle point (C, R) satisfies the conditions of Lemma 4. Equations 4, 5 and 6 imply that C is a convex combination of the elements of $\mathcal{Q}$.

If $R(i) > 0$, then Equation 8 implies that $\mu(i) = 0$. Substituting into Equation 3 yields

$$\mathbb{D}_{\mathrm{KL}}\left[Q_i \,\|\, C\right] = \lambda.$$

If $R(i) = 0$, then Equation 7 implies that $\mu(i) \geq 0$. Combining this with Equation 3 gives

$$\mathbb{D}_{\mathrm{KL}}\left[Q_i \,\|\, C\right] \leq \lambda.$$

For any probability distribution C', we have

$$\begin{aligned}
\mathbb{D}_{\mathrm{KL}}\left[C \,\|\, C'\right] &= \sum_i R(i) \left(\mathbb{D}_{\mathrm{KL}}\left[Q_i \,\|\, C'\right] - \mathbb{D}_{\mathrm{KL}}\left[Q_i \,\|\, C\right]\right) \\
&= \sum_i \mathbb{D}_{\mathrm{KL}}\left[Q_i \,\|\, C'\right] R(i) - \lambda \\
&\leq \mathbb{S}\left[C'\right] - \lambda. \qquad (15)
\end{aligned}$$

Now suppose that C' is also a central distribution of $\mathcal{Q}$ for X. Since C and C' are both central distributions,

$$\mathbb{S}\left[C'\right] = \mathbb{S}\left[C\right] = \lambda.$$

Therefore, by non-negativity of the KL divergence, we have

$$\mathbb{D}_{\mathrm{KL}}\left[C \,\|\, C'\right] = 0,$$

which implies C and C' are identical probability distributions.

Thus, we have proved that $\mathcal{Q}$ has a unique central distribution satisfying the conditions of the theorem. We now prove the "if" direction. Assume that the conditions of the theorem hold for C. Since C is a convex combination of the elements of $\mathcal{Q}$ it is a probability distribution over X. By defining

$$\mu(i) = \lambda - \mathbb{D}_{\mathrm{KL}}\left[Q_i \,\|\, C\right],$$

we have $\mu(i) \geq 0$ and $\mu(i)R(i) = 0$ for every i. We can verify that C and R satisfy the conditions of Lemma 4 and therefore (C, R) is a saddle point. By Lemma 3 C is a central distribution of $\mathcal{Q}$ for X. □

Corollary 3.1. *Let C be the central distribution of $\mathcal{Q}$ for X. For any probability distribution P, we have*

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{Q}(X) \,\|\, P(X)\right] \leq \mathbb{S}_{\mathcal{Q}}\left[P(X)\right] - \mathbb{S}_{\mathcal{Q}}\left[C(X)\right].$$

Proof. By Theorem 3, C is unique, and therefore

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{Q} \,\|\, P\right] = \mathbb{D}_{\mathrm{KL}}\left[C \,\|\, P\right].$$

The result then follows immediately from Equation 15. □

Theorem 4. *The central distribution of $\mathcal{Q}$ for X is a maximizer of the function g defined in Lemma 5.*

Proof. By Lemmas 2 and 3, there exists a saddle point (C, R) of f, where C is the unique central distribution of $\mathcal{Q}$ for X (Theorem 3). Defining $C(X, I) = Q_I(X)R(I)$, Lemma 5 implies that C is a maximizer of the function g. $\square$

Corollary 4.1. *Let C be the central distribution of $\mathcal{Q}$ for X, and let P be a convex combination of the elements of $\mathcal{Q}$, i.e.,*

$$P(x) = \sum_i Q_i(x)R(i),$$

where R is a probability distribution. Then there exists an index k in the support of R such that

$$\mathbb{S}_{\mathcal{Q}}\left[C(X)\right] \geq \mathbb{D}_{\mathrm{KL}}\left[Q_k(X) \,\|\, P(X)\right].$$

Proof. We proceed by contradiction. Suppose that for every k in the support of R, we have $\mathbb{D}_{\mathrm{KL}}\left[Q_k \,\|\, P\right] > \mathbb{S}\left[C\right]$. That implies

$$g(P) > \mathbb{S}\left[C\right] = g(C),$$

where g is the function defined in Lemma 5. Note that $g(P)$ is defined, because P is a convex combination of the elements of $\mathcal{Q}$.

However, by Theorem 4, the distribution C maximizes g, which contradicts the inequality $g(P) > g(C)$. $\square$

2.2 Approximate Central Distributions

Suppose we want to approximate a central distribution of a hierarchy of distributions $\mathcal{H}$. To do so, we need a measure that quantifies the quality of an approximation. A natural choice is the central divergence introduced in Definition 2.3. Since central distributions are intended to serve as representatives of hierarchies of distributions, central divergence can be viewed as a measure of how well a probability distribution represents a hierarchy.

At first glance, computing this quantity appears to require explicit knowledge of the central distributions of $\mathcal{H}$. However, it turns out that the central divergence admits an upper bound that can be estimated without explicitly knowing the central distribution.

In variational settings, it is often more convenient to work with unnormalized measures rather than normalized distributions. Therefore, we state the next theorem in terms of unnormalized measures. We assume throughout that $\tilde{P}(x) > 0$ whenever $Q(x) > 0$ for some $Q \in \mathcal{Q}$, so that all logarithms are well-defined.

Theorem 5. *Let $\mathcal{Q}$ be a finite set of probability distributions over a discrete random variable X. Let $\tilde{P}$ be a linear combination of elements of $\mathcal{R} \subseteq \mathcal{Q}$ with strictly positive weights, and let P be the corresponding normalized distribution. Then*

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{Q} \,\|\, P\right] \leq \sup_{Q \in \mathcal{R}} \mathbb{F}\left[Q, \tilde{P}\right] - \inf_{Q \in \mathcal{Q}} \mathbb{F}\left[Q, \tilde{P}\right],$$

where

$$\mathbb{F}\left[Q, \tilde{P}\right] = \mathbb{H}\left[Q(X)\right] + \mathbb{E}_{Q}\left[\ln \tilde{P}(X)\right].$$

Proof. By Corollary 3.1, we have

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{Q} \,\|\, P\right] \leq \mathbb{S}_{\mathcal{Q}}\left[P\right] - \mathbb{S}_{\mathcal{Q}}\left[C\right].$$

Since P is a convex combination of elements of $\mathcal{R}$ with strictly positive weights, Corollary 4.1 implies that

$$\mathbb{S}_{\mathcal{Q}}\left[C\right] \geq \inf_{Q \in \mathcal{R}} \mathbb{D}_{\mathrm{KL}}\left[Q \,\|\, P\right].$$

Combining these two inequalities yields

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{Q} \,\|\, P\right] \leq \mathbb{S}_{\mathcal{Q}}\left[P\right] - \inf_{Q \in \mathcal{R}} \mathbb{D}_{\mathrm{KL}}\left[Q \,\|\, P\right]. \tag{16}$$

Let $\tilde{P}(X) = ZP(X)$, where $Z > 0$ is a normalization constant. Then for any Q,

$$\mathbb{D}_{\mathrm{KL}}\left[Q \,\|\, P\right] = \ln Z - \mathbb{F}\left[Q, \tilde{P}\right].$$

Since $\ln Z$ does not depend on Q, we have

$$\inf_{Q \in \mathcal{R}} \mathbb{D}_{\mathrm{KL}}\left[Q \,\|\, P\right] = \ln Z - \sup_{Q \in \mathcal{R}} \mathbb{F}\left[Q, \tilde{P}\right]$$
$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q \,\|\, P\right] = \ln Z - \inf_{Q \in \mathcal{Q}} \mathbb{F}\left[Q, \tilde{P}\right],$$

and the result follows from Equation 16. □

Since central distributions are defined through marginal distributions, their construction may be viewed as combining prior specification and Bayesian inference within a single optimization framework. Given n observations, one may perform inference using the central distribution of the corresponding finite sequence S_n, without separately specifying a prior distribution.

As discussed later, in the case of sequences of random variables, central distributions are defined on the underlying infinite sequence. Consequently, the central distribution of a finite initial segment S_n should be viewed as an approximation to the corresponding marginal of the infinite sequence central distribution. A natural question is how accurate this approximation is.

2.3 Parametric Families and Central Priors

The results of the previous subsection were established under the assumption that $\mathcal{Q}$ is finite. Many of the underlying constructions, however, can be extended to parametric families in which $\mathcal{Q}$ is indexed by a finite-dimensional parameter vector Θ:

$$\mathcal{Q} = \{Q_\theta \mid \theta \in \mathrm{val}(\Theta)\}\,.$$

In this setting, Θ may be regarded as a random variable, with each Q_θ interpreted as a conditional distribution $Q(x \mid \theta)$. The weight function $R(i)$ appearing in the finite case is then replaced by a prior distribution over Θ.

A rigorous development of this extension requires additional measure-theoretic and regularity assumptions and is beyond the scope of the present paper.

Definition 2.6 (Central Prior)**.** Let X be a random variable, and let $\mathcal{Q}$ be a set of probability distributions over X, indexed by a parameter Θ. A nonnegative measure c on Θ is called a central prior of $\mathcal{Q}$ for X if,

$$C(x) = \int Q(x \mid \theta)c(\theta)d\theta$$

is a central distribution of $\mathcal{Q}$ for X.

Definition 2.7 (Interior Distribution)**.** Let $\mathcal{Q}$ be a set of probability distributions over a random variable X. A distribution $R \in \mathcal{Q}$ is called an *interior distribution* of $\mathcal{Q}$ if and only if the sets $\mathcal{Q}$ and $\mathcal{Q} \setminus \{R\}$ have the same central distributions for X.

Theorem 6. *Let X be a discrete random variable with finite support, and let $\mathcal{Q}$ be a finite set of probability distributions over X. Let C be the central distribution of $\mathcal{Q}$ for X. If for some $Q_k \in \mathcal{Q}$*

$$\mathbb{D}_{\mathrm{KL}}\left[Q_k \,\|\, C\right] < \mathbb{S}_{\mathcal{Q}}\left[C\right], \tag{17}$$

then Q_k is an interior distribution of $\mathcal{Q}$.

Proof. By Theorem 3, C is a convex combination of elements of $\mathcal{Q}$, and Equation 17 implies that $R(k) = 0$. Hence, C is also a convex combination of elements of $\mathcal{Q} \setminus \{Q_k\}$ and the conditions of Theorem 3 remain satisfied for C and $\mathcal{Q} \setminus \{Q_k\}$. Thus, C is the unique central distribution of $\mathcal{Q} \setminus \{Q_k\}$ and $\mathcal{Q}$ for X. □

Theorem 7. *Let X be a random variable and let $\mathcal{Q}$ be a set of probability distributions over X. If $R \in \mathcal{Q}$ can be expressed as a convex combination of elements of $\mathcal{Q} \setminus \{R\}$, then R is an interior distribution of $\mathcal{Q}$.*

Proof. Let $R = \sum_i \alpha_i Q_i$, where $Q_i \in \mathcal{Q} \setminus \{R\}$, $\alpha_i \geq 0$, and $\sum_i \alpha_i = 1$.

Fix an arbitrary probability distribution P. By the convexity of the KL divergence in its first argument, we have

$$\mathbb{D}_{\mathrm{KL}}\left[R \,\|\, P\right] \leq \sum_i \alpha_i \mathbb{D}_{\mathrm{KL}}\left[Q_i \,\|\, P\right].$$

Note that the right-hand side is a weighted average of the KL divergences. So, there must be at least one index k such that

$$\mathbb{D}_{\mathrm{KL}}\left[R \,\|\, P\right] \leq \mathbb{D}_{\mathrm{KL}}\left[Q_k \,\|\, P\right].$$

Since $Q_k \in \mathcal{Q} \setminus \{R\}$, it follows that

$$\mathbb{S}_{\mathcal{Q}}\left[P\right] = \mathbb{S}_{\mathcal{Q}\setminus\{R\}}\left[P\right]$$

for every probability distribution P. Therefore, $\mathcal{Q}$ and $\mathcal{Q} \setminus \{R\}$ have the same central distributions. $\square$

One important consequence of this theorem is that, in order to identify a central distribution within a parametric model, it suffices to restrict attention to the set of likelihood distributions induced by different values of the model parameters.

Theorem 8. *Let X and T be two random variables, and let $\mathcal{Q}$ be a set of distributions over the pair (X, T). Assume that for every $Q \in \mathcal{Q}$, we have*

$$Q(X, T) = R(X \mid T)Q(T),$$

where R is a fixed conditional distribution. If C is a central distribution of $\mathcal{Q}$ for T, then $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X, T).

Proof. Let C be a central distribution of $\mathcal{Q}$ for T. For any arbitrary probability distribution P, the definition of a central distribution implies

$$\mathbb{S}\left[R(X \mid T)C(T)\right] \leq \mathbb{S}\left[R(X \mid T)P(T)\right]. \tag{18}$$

Since the KL divergence is nonnegative, for any probability distribution P and every $Q \in \mathcal{Q}$ we have

$$\sum_t \mathbb{D}_{\mathrm{KL}}\left[R(X \mid t) \,\|\, P(X \mid t)\right] Q(t) \geq 0.$$

That implies

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X, T) \,\|\, R(X \mid T)P(T)\right] \leq \mathbb{D}_{\mathrm{KL}}\left[Q(X, T) \,\|\, P(X, T)\right].$$

Taking the supremum over $Q \in \mathcal{Q}$ preserves the inequality, since it holds pointwise for every $Q \in \mathcal{Q}$:

$$\mathbb{S}\left[R(X \mid T)P(T)\right] \leq \mathbb{S}\left[P(X,T)\right].$$

Combining this inequality with Equation 18, we conclude that $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X,T). $\square$

The primary application of this theorem is to derive central distributions for variables of interest from central distributions of sufficient statistics.

The following theorem provides a sufficient fixed-point characterization of central priors and highlights a formal connection between the central distribution framework and Bernardo's reference prior methodology [3].

Theorem 9. *Let X be a discrete random variable, and let $\mathcal{Q}$ be a set of probability distributions over X, indexed by a finite-dimensional vector Θ. Let c be a (possibly improper) prior distribution for Θ such that*

$$c(\theta) \propto \exp\left\{\sum_x Q(x \mid \theta) \ln c(\theta \mid x)\right\}. \tag{19}$$

Then, c is a central prior of $\mathcal{Q}$ for X.

Proof. Since $Q_\theta \in \mathcal{Q}$ is a probability distribution, for every θ there exists x such that $Q_\theta(x) > 0$. Equation 19 therefore implies $c(\theta \mid x) > 0$, and consequently $c(\theta) > 0$.

Using Equation 19 and $c(\theta) > 0$, we obtain

$$\begin{aligned} c(\theta) &= \exp\left\{-\lambda + \sum_x Q(x \mid \theta) \ln \frac{Q(x \mid \theta)c(\theta)}{C(x)}\right\} \\ &= \exp\left\{-\lambda + \mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right] + \ln c(\theta)\right\} \\ &= c(\theta)\exp\left\{-\lambda + \mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right]\right\}, \end{aligned}$$

and it follows that

$$\mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right] = \lambda.$$

Since this holds for every θ, and λ is constant, a parametric extension of Theorem 3 implies that C, the marginal distribution induced by c, is a central distribution of $\mathcal{Q}$ for X. Therefore, by Definition 2.6, c is a central prior of $\mathcal{Q}$ for X. $\square$

This theorem is proved using Theorem 3. However, Theorem 4 suggests an alternative route to establishing the connection with reference priors. Since reference priors are defined by maximizing missing information, the optimization characterization provided by Theorem 4 also offers another perspective on this relationship.

2.4 Asymptotic Central Distributions

Many probabilistic models are naturally formulated in terms of a sequence of random variables whose length is not fixed in advance. For example, when a coin is tossed repeatedly, there is no natural upper bound on the number of tosses, and any finite sample can be viewed as an initial segment of an infinite sequence.

Motivated by such examples, let S_n denote the first n variables of an infinite sequence of random variables:

$$S_n = (X_1, X_2, \ldots, X_n).$$

The collection $\{S_n\}_{n=1}^{\infty}$ therefore defines a second sequence of random variables.

In many settings, our uncertainty about S_n can be represented by a central distribution. A prominent example arises when the sequence is assumed to be exchangeable. As discussed earlier, exchangeability can be viewed as a form of partial knowledge that defines a set of plausible distributions, from which a central distribution for S_n can be constructed.

However, this central distribution generally depends on the sequence length n. In this work, we adopt the principle that the probability measure used to represent prior beliefs about the environment should depend only on the available information about the environment itself, and not on arbitrary aspects of the experimental setup, such as the number of variables under consideration.

Importantly, a distribution defined over a longer sequence induces marginal distributions over all shorter initial segments. This suggests that, rather than constructing a separate central distribution for each S_n, we should seek a single probability measure over the infinite sequence, from which the distributions of all finite initial segments can be obtained as marginals.

This motivates the construction of a central distribution that is well defined over S_∞.

Definition 2.8 (Asymptotic Central Distribution)**.** Let $\{S_n\}_{n=1}^{\infty}$ be a sequence of random variables and $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each S_n. Let C be also a probability measure that defines a distribution over every S_n. C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$, if and only if we have

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{C}}\left[\mathcal{Q}(S_n) \,\|\, C(S_n)\right] = 0.$$

Intuitively, this definition states that as $n \to \infty$ the distribution that C induces over S_n becomes increasingly similar to a central distribution for S_n. In other words, C is a central distribution for the infinite sequence S_∞.

Our next goal is to extend Theorem 3 so that it also applies in this asymptotic setting.

Theorem 10 (Asymptotic Centrality)**.** *Let $\{S_n\}_{n=1}^{\infty}$ be a sequence of discrete random variables with finite support and let $\mathcal{Q}$ be a finite set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each S_n. Suppose that C is a convex combination of the elements of $\mathcal{Q}$. If there exists a sequence of constants $\{\lambda_n\}_{n=1}^{\infty}$ such that*

$$\lim_{n\to\infty} \sup_{Q\in\mathcal{Q}} \left|\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right| = 0,$$

then C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$.

Proof. For every n, let C_n be the unique central distribution of $\mathcal{Q}$ for S_n, and let μ_n denote the corresponding constant from Theorem 3. Since C is a convex combination of elements of $\mathcal{Q}$, $C(S_n)$ is well-defined and Corollary 4.1 implies

$$\mu_n \geq \inf_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right],$$

and, by the definition of a central distribution,

$$\mu_n \leq \mathbb{S}\left[C(S_n)\right].$$

Moreover, we have

$$\left|\inf_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right| \leq \sup_{Q\in\mathcal{Q}} \left|\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right|,$$

and

$$\left|\mathbb{S}\left[C(S_n)\right] - \lambda_n\right| \leq \sup_{Q\in\mathcal{Q}} \left|\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right|.$$

Hence, both $\{\inf_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\}_{n=1}^{\infty}$ and $\{\mathbb{S}\left[C(S_n)\right] - \lambda_n\}_{n=1}^{\infty}$ converge to 0. By the squeeze theorem, the sequence $\{\mu_n - \lambda_n\}_{n=1}^{\infty}$ also converges to 0. We may write

$$\begin{aligned}\lim_{n\to\infty} \left(\mathbb{S}\left[C(S_n)\right] - \mu_n\right) &= \lim_{n\to\infty} \left(\mathbb{S}\left[C(S_n)\right] - \lambda_n\right) - \lim_{n\to\infty} \left(\mu_n - \lambda_n\right) \\ &= 0.\end{aligned}$$

By Corollary 3.1, we have

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{Q}(S_n) \,\|\, C(S_n)\right] \leq \mathbb{S}\left[C(S_n)\right] - \mu_n,$$

and the squeeze theorem implies

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{C}}\left[\mathcal{Q}(S_n) \,\|\, C(S_n)\right] = 0.$$

Thus, by Definition 2.8, C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$. $\square$

Theorem 10 provides a sufficient condition for asymptotic centrality, while general existence and uniqueness results for infinite sequences remain open.

We present the following construction as a conjectured framework. Under standard regularity conditions, this approach motivates the functional form of central priors and serves as a conceptual bridge to stimulate future formal proofs. This construction is motivated by a similar asymptotic approach in [3].

Conjecture 1. *Let $\{T_n\}_{n=1}^{\infty}$ be a sequence of random variables, and let $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each T_n. Assume that $\mathcal{Q}$ is indexed by a finite-dimensional vector Θ, and $\{T_n\}_{n=1}^{\infty}$ is a consistent sequence of estimators for Θ. Define*

$$c(\theta) \propto \lim_{n\to\infty} \frac{P(\theta \mid T_n = t_n)|_{t_n=\theta}}{\mu_n}, \tag{20}$$

where $\{\mu_n\}_{n=1}^{\infty}$ is a sequence of positive constants and $P(\theta \mid T_n)$ denotes the posterior distribution obtained from an arbitrary smooth positive prior. If this limit exists and the convergence is uniform over θ, under the usual regularity conditions, $c(\theta)$ is a central prior of $\mathcal{Q}$ for $\{T_n\}_{n=1}^{\infty}$.

Let k denote the normalizing constant converting proportionality into equality. Since the convergence is uniform and the logarithm is a continuous function, Equation 20 implies

$$\lim_{n\to\infty} \sup_{\theta} \left| \ln \frac{P(\theta \mid T_n = t_n)|_{t_n=\theta}}{c(\theta)} - \ln \mu_n + \ln k \right| = 0.$$

Because T_n is a consistent estimator of every θ, the sampling distribution $Q\,(T_n \mid \theta)$ becomes increasingly concentrated around θ as $n \to \infty$. Therefore, we may write

$$\lim_{n\to\infty} \sup_{\theta} \left| \sum_{t_n} Q(t_n \mid \theta) \ln \frac{P(\theta \mid t_n)}{c(\theta)} - \ln \mu_n + \ln k \right| = 0.$$

Furthermore, under the usual regularity conditions, the posterior distribution becomes asymptotically insensitive to the choice of smooth positive prior. Motivated by this, we heuristically replace $P(\theta \mid T_n)$ with $c(\theta \mid T_n)$, obtaining

$$\lim_{n\to\infty} \sup_{\theta} \left| \sum_{t_n} Q(t_n \mid \theta) \ln \frac{Q(t_n \mid \theta)}{C(t_n)} - \ln \mu_n + \ln k \right| = 0,$$

where

$$C(t_n) = \int Q(t_n \mid \theta) c(\theta) d\theta.$$

Letting $\lambda_n = \ln \mu_n - \ln k$, we observe that

$$\lim_{n\to\infty} \sup_{\theta} \left| \mathbb{D}_{\mathrm{KL}} \left[Q\left(T_n \mid \theta\right) \| C(T_n) \right] - \lambda_n \right| = 0.$$

By a parametric extension of Theorem 10, it can be shown that C is a central distribution of $\mathcal{Q}$ for $\{T_n\}_{n=1}^{\infty}$ and therefore $c(\theta)$ is a central prior for the estimator sequence $\{T_n\}_{n=1}^{\infty}$.

Example 2.2 (Binomial Experiment)**.** Consider a binomial experiment consisting of n independent Bernoulli trials. Let $\mathcal{Q}$ denote the family of probability distributions corresponding to different values of the success probability $\theta \in [0, 1]$, with θ serving as the indexing parameter for $\mathcal{Q}$. Let T_n be the sample proportion of successes in the n trials. Since T_n is a consistent estimator of θ, we can apply Equation 20 to determine a central distribution for the sequence $\{T_n\}_{n=1}^{\infty}$.

Assuming a uniform prior on θ, Bayes' rule gives the posterior distribution

$$P\left(\theta \,\middle|\, T_n = \tfrac{k}{n}\right) = (n+1)\binom{n}{k}\theta^k(1-\theta)^{n-k},$$

which is a Beta$(k+1, n-k+1)$ distribution. Using Stirling's formula, we derive its asymptotic form:

$$P\left(\theta \,\middle|\, T_n = \tfrac{k}{n}\right) \sim \frac{(n+1)n^{(n+\frac{1}{2})}}{\sqrt{2\pi}(n-k)^{(n-k+\frac{1}{2})}k^{(k+\frac{1}{2})}}\,\theta^k(1-\theta)^{n-k}.$$

Evaluating the asymptotic expression at $\frac{k}{n} = \theta$, we obtain

$$P(\theta \mid T_n = \frac{k}{n})|_{\frac{k}{n}=\theta} \sim \frac{n+1}{\sqrt{2\pi n}}\,\theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}.$$

Therefore, if we define

$$\mu_n = \frac{n+1}{\sqrt{2\pi n}},$$

the following limit exists and the convergence is uniform over $\theta \in (0, 1)$:

$$\lim_{n\to\infty} \frac{P(\theta \mid T_n = t_n)|_{t_n=\theta}}{\mu_n} = \theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}.$$

A central distribution over an estimator sequence cannot, by itself, be used to perform inference at the level of observed outcomes. To do so, we must convert it into a distribution over the primary variables of interest. Theorem 8 provides a simple method for this conversion.

Example 2.3 (Fused Multinomial Model)**.** Let $\{S_n\}_{n=1}^{\infty}$ be a sequence in which each $S_n = (X_1, X_2, \ldots, X_n)$ represents n independent trials of an experiment with k possible outcomes, and $\mathcal{Q}$ be a set of probability distributions over S_n indexed by a real vector θ. Let $f : \mathcal{X} \to \mathcal{Y}$ be a deterministic function that maps each outcome of X_i to a label $Y_i = f(X_i)$, where $\mathcal{Y} = \{1, \ldots, m\}$. Assume that for all i and $Q_\theta \in \mathcal{Q}$, if $f(x_i) = f(x_i')$ we have

$$Q_\theta(x_1, x_2, \ldots, x_n) = Q_\theta(x_1', x_2', \ldots, x_n').$$

That is, the conditional distribution $Q_\theta(X_i \mid Y_i = y)$ is uniform over the preimage $f^{-1}(y)$.

We consider $\theta = (\theta_1, \ldots, \theta_m)$ to be a probability vector over $\mathcal{Y}$, where each θ_y specifies the probability of observing label y. For a sequence S_n, let n_y denote the number of times label $y \in \mathcal{Y}$ appears. Let $|f^{-1}(y)|$ denote the number of outcomes in $\mathcal{X}$ that are mapped to label y by f. Then the probability of observing S_n under $Q_\theta \in \mathcal{Q}$, denoted $Q(S_n \mid \theta)$, is:

$$Q(S_n \mid \theta) = \prod_{y=1}^{m} \left(\frac{1}{|f^{-1}(y)|} \cdot \theta_y \right)^{n_y}.$$

Consider the estimator sequence $T_n = \frac{1}{n}(n_1, n_2, \ldots, n_m)$, which is a consistent estimator of θ. The conditional distribution $Q_\theta(S_n \mid T_n)$ is given by:

$$R(S_n \mid T_n) = \frac{1}{n!} \prod_{y=1}^{m} \frac{n_y!}{|f^{-1}(y)|^{n_y}}. \tag{21}$$

Note that this distribution is independent of θ. Therefore, a central distribution for S_n can be obtained from a central distribution for T_n.

As $n \to \infty$, the distribution of T_n converges to a multivariate Gaussian distribution. Using this fact and Equation 20, we can determine a central prior for $\{T_n\}_{n=1}^{\infty}$:

$$c(\theta) \propto \prod_{y=1}^{m} \theta_y^{-\frac{1}{2}}.$$

This is a Dirichlet prior and induces a Dirichlet-multinomial distribution for each T_n:

$$C(T_n) = \frac{\Gamma\left(\frac{m}{2}\right)\Gamma(n+1)}{\Gamma\left(n + \frac{m}{2}\right)} \prod_{y=1}^{m} \frac{\Gamma\left(n_y + \frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)\Gamma(n_y + 1)}.$$

Here, C is a central distribution for the sequence $\{T_n\}_{n=1}^{\infty}$, viewed as a probability measure on the infinite sequence T_∞. The notation $C(T_n)$ denotes the marginal distribution of T_n.

By Theorem 8 and Equation 21, we can derive a central distribution for $\{S_n\}_{n=1}^{\infty}$, which induces the following distribution over S_n:

$$C(S_n) = \frac{\Gamma\left(\frac{m}{2}\right)}{\Gamma\left(n + \frac{m}{2}\right)} \prod_{y=1}^{m} \frac{\Gamma\left(n_y + \frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right) |f^{-1}(y)|^{n_y}}.$$

When dealing with layered ignorance and seeking the central distribution of a hierarchy of distributions, we typically encounter situations in which a given level of the hierarchy contains only finitely many alternatives. In such cases, and under appropriate regularity conditions, if there exists a consistent estimator for the index identifying these alternatives, then the central distribution reduces to an equally weighted average of the corresponding representative central distributions.

3 Illustrative Example: Attribute Selection

A natural application of hierarchical central distributions arises in feature selection. When the relevant set of features is unknown, each candidate feature set induces a distinct family of admissible distributions. Feature selection can therefore be viewed as uncertainty over a hierarchy of probabilistic models, making it a natural setting for the central distribution framework.

To illustrate this idea , we provide a proof of concept experiment using synthetic data. The purpose of this section is not to establish a competitive feature selection procedure, but to illustrate how hierarchical central distributions can be used to represent uncertainty over model structure. A comprehensive empirical evaluation against established feature selection methods, such as Spike-and-Slab priors, on real-world datasets lies beyond the scope of this foundational paper and is left for future work.

We consider a classification problem where the goal is to predict a target class, denoted by Y, from a set of observed nominal attributes $(A_1, A_2, \ldots, A_n)$. We assume that Y may depend on an unknown subset of the attributes, rather than the full set.

To capture this relationship, we use the fused multinomial model introduced in Section 2.4. As an example, assume that Y depends only on A_2 and A_5. Then, we define the labeling function f as follows:

$$f(A_1, A_2, \ldots, A_n, Y) = (A_2, A_5, Y).$$

In a fused multinomial model with this labeling function, the target class Y is conditionally independent of all other attributes given A_2, A_5.

For every possible subset of attributes, a distinct fused multinomial model of this form is constructed. Each such model corresponds to a set of plausible probability distributions. Collecting these sets yields our overall uncertainty representation, which forms a one-level hierarchy of distributions (Definition 2.2).

Let $C_{\mathcal{H}^*}$ denote a central distribution of the full hierarchy, and let C_{f_i} denote the central distribution of the fused multinomial model associated with the labeling function f_i. As discussed earlier, because the number of candidate models is finite, we have

$$C_{\mathcal{H}^*}(S_n) \propto \sum_i C_{f_i}(S_n). \tag{22}$$

To justify this, observe that we can construct a consistent estimator for the model indicator I by examining equality relationships among the observed frequencies of outcomes. Specifically, the estimator identifies outcomes that occur with equal frequencies and returns the indicator corresponding to the model that supports those equalities.

The optimal Bayesian decision rule depends on the choice of loss function. In this example, we use the zero–one loss, under which the optimal strategy predicts the target class with the highest posterior probability. These posterior probabilities are computed using the central distribution of Equation 22. It turns out that each subset-specific fused multinomial model is assigned its corresponding Jeffreys' prior. Consequently, the resulting predictive distribution can be interpreted as Bayesian model averaging over all candidate feature subsets, with each model equipped with its Jeffreys' prior. The central distribution framework provides a minimax-KL interpretation for this aggregation.

To see how this approach works in practice, we carried out a synthetic experiment with the following setup. We considered binary attributes and generated each attribute independently with equal probability for 0 and 1. The target class was defined as a deterministic function of a fixed subset of 3 out of 12 total attributes, as shown in Table 1. The same subset of relevant attributes and class assignment function were used across all trials.

For each experiment, we generated a test set of size 100. A larger test set was used to reduce variability in performance estimates, taking advantage of the synthetic nature of the data. To assess robustness to irrelevant features, we progressively removed one attribute at a time from the full set of 12, starting with irrelevant ones and stopping just before removing any of the three relevant attributes.

Table 1: Mappings of 3-bit inputs under class functions Y_1 and Y_2

Input (bits)	Y_1(Input)	Y_2(Input)
000	a	A
001	b	B
010	c	A
011	d	C
100	e	D
101	f	D
110	g	D
111	h	E

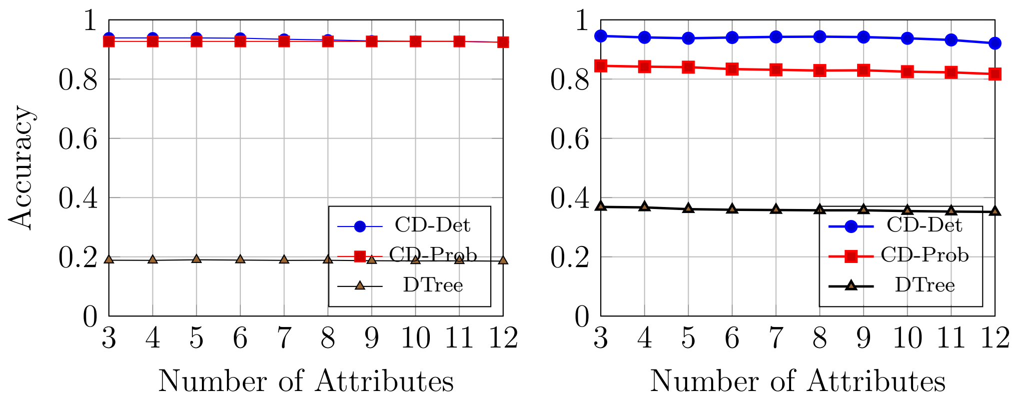

Figure 1: Accuracy versus number of attributes for 20 training samples. Left: under class assignment function Y_1. Right: under class assignment function Y_2.

At each step, we trained and tested two versions of the central distribution: one assuming that the target class is a deterministic function of the attributes (CD-Det), and one allowing the target class to follow a conditional probability distribution given the attributes (CD-Prob). Both versions are constructed by aggregating over 2^{12} fused multinomial models, corresponding to all subsets of the 12 attributes. As a baseline, we also trained a decision tree classifier (DTree). All methods were trained on the same training data and evaluated on the same test data. Each configuration was repeated for 100 independent trials, and we reported the average accuracy (correct predictions divided by total test instances) over these trials.

The results, presented in Figure 1, show that both central distribution methods substantially outperform the decision tree on the two classification tasks. Interestingly, the relative difficulty of the tasks differs between the

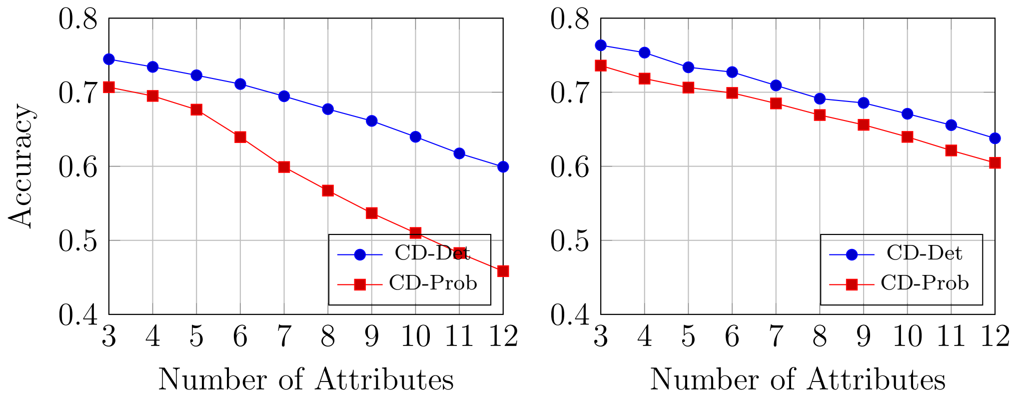

Figure 2: Accuracy versus number of attributes for 10 training samples. Left: under class assignment function Y_1. Right: under class assignment function Y_2.

methods. For the central distribution approach, Y_2 is more challenging than Y_1, resulting in a larger gap between the deterministic and probabilistic variants. This difficulty arises from strong non-deterministic dependencies between subsets of input bits and the assigned class, which can create misleading probabilistic associations. In contrast, the decision tree performs worse on Y_1 than on Y_2. Overall, the deterministic central distribution achieved the highest accuracy, while the probabilistic version produced similar but slightly weaker results.

With 20 training samples, removing irrelevant attributes had little effect on performance. Accuracy remained nearly constant as the number of attributes decreased from 12 to 3, and the decision tree showed a similar insensitivity to attribute count. However, with only 10 training samples (Figure 2), accuracy increased noticeably as irrelevant attributes were removed, indicating that irrelevant features become more problematic in low-data regimes.

The implementation used in this experiment evaluates all 2^{12} candidate feature subsets and therefore remains computationally feasible only because the dimensionality is small. More generally, because the central distribution aggregates over all candidate subsets, the resulting predictor is not expected to scale to high-dimensional problems without additional approximation techniques. Developing scalable approximation methods could be a direction for future work.

4 Conclusion

We introduced the central distribution, a minimax-KL representative of a set or hierarchy of probability distributions. This provides an information-theoretic framework for converting set-valued and hierarchical uncertainty into a single distribution suitable for Bayesian inference.

We established existence, uniqueness, and optimization-theoretic characterizations of central distributions, introduced central divergence as a measure of representational quality, and determined bounds for approximating central distributions. We further developed the corresponding notion of a central prior and identified a sufficient fixed-point condition linking central priors to Bernardo's reference prior methodology. We also proposed an asymptotic extension of the framework for future investigation.

The framework also provides a natural way to represent uncertainty about model structure through hierarchical collections of distributions. As a proof of concept, we illustrated this idea in a feature selection setting.

Several important questions remain open. In particular, a rigorous extension of the theory to general parametric families, computational methods for computing and approximating central distributions, and a rigorous asymptotic theory for central priors require further investigation. It would also be valuable to study central distributions in a broader range of practical statistical problems in order to better understand their computational and inferential properties. We hope that the central distribution framework provides a useful foundation for the study of Bayesian inference and hierarchical uncertainty from an information-theoretic perspective.

References

[1] Sheldon Jay Axler. *Linear Algebra Done Right*. Undergraduate Texts in Mathematics. Springer, New York, 2023.

[2] Jose M. Bernardo. Reference posterior distributions for bayesian inference. *Journal of the Royal Statistical Society: Series B (Methodological)*, 41(2):113–128, 01 1979.

[3] José M. Bernardo. Reference analysis. In D.K. Dey and C.R. Rao, editors, *Bayesian Thinking*, volume 25 of *Handbook of Statistics*, pages 17–90. Elsevier, 2005.

[4] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

[5] E. T. Jaynes. Information theory and statistical mechanics. *Phys. Rev.*, 106:620–630, May 1957.

[6] Harold Jeffreys. An invariant form for the prior probability in estimation problems. *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences*, 186(1007):453–461, 09 1946.

[7] Hidetoshi Komiya. Elementary proof for Sion's minimax theorem. *Kodai Mathematical Journal*, 11(1):5 – 7, 1988.

[8] S. Kullback and R. A. Leibler. On Information and Sufficiency. *The Annals of Mathematical Statistics*, 22(1):79 – 86, 1951.

[9] David Rios Insua and Fabrizio Ruggeri. *Robust Bayesian Analysis*, volume 152 of *Lecture Notes in Statistics*. Springer-Verlag, New York, 2000.

[10] P. Walley. *Statistical Reasoning with Imprecise Probabilities*. Chapman & Hall/CRC Monographs on Statistics & Applied Probability. Taylor & Francis, 1991.

A Expectation as a Linear Map

While studying linear algebra, one quickly encounters the idea that linear maps often reveal the essential structure of a mathematical theory. Probability theory provides a familiar example: expectation is a linear operation. This observation motivates a simple question: how much of elementary probability theory can be reconstructed if expectation, rather than probability, is taken as the starting point? In this note, we explore this question by treating expectation as a positive normalized linear functional on a space of actions.

Definition A.1 (Sample Space)**.** A sample space, denoted by $\mathcal{S}$, is the set of all possible outcomes of a random experiment. That is, the outcome of a random experiment can *always* be represented by *exactly one* element of $\mathcal{S}$.

Definition A.2 (Action Space)**.** Let $\mathcal{S}$ be a sample space. The action space of $\mathcal{S}$, denoted by $\mathcal{A}_{\mathcal{S}}$, is the set of all functions from $\mathcal{S}$ to $\mathbb{R}$, where $\mathbb{R}$ is the field of real numbers.

Proposition. The action space equals $\mathbb{R}^{\mathcal{S}}$, and is a vector space.

We can think of the members of $\mathcal{A}_{\mathcal{S}}$ as hypothetical actions or gambles. Each action associates an amount of loss with every possible outcome $s \in \mathcal{S}$. This loss is quantified by a real number. A negative loss indicates profit.

Definition A.3 (Sure Thing)**.** The *sure thing*, denoted by c, is an element of $\mathcal{A}_{\mathcal{S}}$, such that $c(s) = 1$ for every $s \in \mathcal{S}$.

Definition A.4. Let $\mathcal{E} \subseteq \mathcal{S}$. The restriction to $\mathcal{E}$ operator, denoted by $R_{\cap\mathcal{E}}$, is an operator on $\mathcal{A}_{\mathcal{S}}$, such that $\ell' = R_{\cap\mathcal{E}}(\ell_{\mathcal{S}})$ for any $\ell_{\mathcal{S}} \in \mathcal{A}_{\mathcal{S}}$ is an element of $\mathcal{A}_{\mathcal{S}}$ and

$$\ell'(s) = \begin{cases} \ell_{\mathcal{S}}(s) & s \in \mathcal{E}, \\ 0 & s \in \mathcal{S} - \mathcal{E}. \end{cases}$$

We denote the range of $R_{\cap\mathcal{E}}$ by $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$.

Proposition. The restriction operator is a linear map.

Proposition. $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ is an invariant subspace of $\mathcal{A}_{\mathcal{S}}$ under $R_{\cap\mathcal{E}}$. If $\{\mathcal{E}_1, \mathcal{E}_2, \ldots\}$ is a partition of $\mathcal{S}$, then

$$\mathcal{A}_{\mathcal{S}} = \mathcal{A}_{\mathcal{S}\cap\mathcal{E}_1} \oplus \mathcal{A}_{\mathcal{S}\cap\mathcal{E}_2} \oplus \ldots,$$

where $\oplus$ indicates direct sum [1].

The restriction operator restricts an action to a subset of the sample space by simply ignoring its payoff for any outcome that is not in that subset.

Definition A.5 (Linear Functional)**.** Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$. A linear functional on $\mathcal{V}$ is a linear map from $\mathcal{V}$ to $\mathbb{F}$.

Definition A.6 (Expectation)**.** An *expectation* for $\mathcal{S}$ is a linear functional on $\mathcal{A}_{\mathcal{S}}$, denoted by $\mathbb{E}_{\mathcal{S}}$, with the following properties:

- For the sure thing $c_{\mathcal{S}}$, we have $\mathbb{E}_{\mathcal{S}}[c_{\mathcal{S}}] = 1$.
- For any $\mathcal{E} \subseteq \mathcal{S}$, we have $\mathbb{E}_{\mathcal{S}}[R_{\cap\mathcal{E}}(c_{\mathcal{S}})] \geq 0$.

Proposition. $\mathbb{E}_{\mathcal{S}}$ is an element of the dual space of $\mathcal{A}_{\mathcal{S}}$.

Definition A.7 (Conditional Expectation)**.** Let $\mathbb{E}_{\mathcal{S}}$ be an expectation for $\mathcal{S}$ and let $\mathcal{E} \subseteq \mathcal{S}$. $\mathbb{E}_{\mathcal{S}}$ conditioned on $\mathcal{E}$, denoted by $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$, is a linear map on $\mathcal{S}$ with the following properties:

- For the sure thing $c_{\mathcal{S}}$, we have $\mathbb{E}_{\mathcal{S}|\mathcal{E}}[c_{\mathcal{S}}] = 1$.
- For any $\ell_{\mathcal{S}} \in \mathcal{A}_{\mathcal{S}}$, and $\mathcal{E} \subseteq \mathcal{S}$, we have $\mathbb{E}_{\mathcal{S}|\mathcal{E}}[\ell_{\mathcal{S}}] = 0$ if and only if $\mathbb{E}_{\mathcal{S}}[R_{\cap\mathcal{E}}(\ell_{\mathcal{S}})] = 0$.

Proposition. The conditional expectation $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$ is only well-defined when $\mathbb{E}_{\mathcal{S}}[R_{\cap\mathcal{E}}(c_{\mathcal{S}})] \neq 0$.

It is important to note that conditioning on $\mathcal{E}$ fundamentally differs from restricting to $\mathcal{E}$. When we restrict an action, we simply ignore its payoffs outside of $\mathcal{E}$. When we condition on $\mathcal{E}$, however, we act with the certain knowledge that $\mathcal{E}$ has actually occurred. As $\mathcal{E}$ becomes increasingly unlikely, the disparity between these two concepts grows, because conditioning forces a rare event to become our new baseline. Yet, a fundamental exception occurs when our expected payoff within $\mathcal{E}$ is zero. No matter how unlikely $\mathcal{E}$ is, knowing that it has occurred cannot transform an expected payoff of zero into a non-zero value. Therefore, we require the null space of the conditional expectation to coincide perfectly with the null space induced by the restriction operator.

At this point, we have enough structure to reason about uncertainty and evaluate actions. Interestingly, we do not need to explicitly define probability. However, in order to construct a more familiar framework, we provide a definition of conventional probability.

Definition A.8 (Probability)**.** We define the probability of $\mathcal{E} \subseteq \mathcal{S}$ as follows:

$$P(\mathcal{E}) = \mathbb{E}_{\mathcal{S}}[R_{\cap\mathcal{E}}(c_{\mathcal{S}})] \cdot$$

Theorem 11. *Definition A.8 satisfies Kolmogorov's axioms of probability.*

We now establish the familiar relationship between conditional expectation and normal expectation.

Lemma 6. *Let $\mathcal{F}$ and $\mathcal{G}$ be two linear functionals defined on a vector space $\mathcal{V}$. $\mathcal{F}$ and $\mathcal{G}$ have the same null space if and only if $\mathcal{F} = \lambda\mathcal{G}$ for some $\lambda \neq 0$.*

Proof. If $\mathcal{F} = \lambda\mathcal{G}$, $\mathcal{F}[v] = 0$ if and only if $\mathcal{G}[v] = 0$ for every $v \in \mathcal{V}$. This proves the if part.

We prove the only if part by contradiction. Assume $\mathcal{F} \neq \lambda\mathcal{G}$. Then there exist $u, v \in \mathcal{V}$ such that

$$\mathcal{F}[u]\,\mathcal{G}[v] - \mathcal{F}[v]\,\mathcal{G}[u] \neq 0\cdot$$

Let $w = \mathcal{F}[u]\,v - \mathcal{F}[v]\,u$. Since $\mathcal{V}$ is a vector space $w \in \mathcal{V}$. $\mathcal{F}$ and $\mathcal{G}$ are linear. Hence, $\mathcal{F}[w] = 0$ and $\mathcal{G}[w] = \mathcal{F}[u]\,\mathcal{G}[v] - \mathcal{F}[v]\,\mathcal{G}[u] \neq 0$ and the null spaces of $\mathcal{F}$ and $\mathcal{G}$ are not equal. This contradiction completes the proof. □

Theorem 12. *Let $\ell_{\mathcal{S}} \in \mathcal{A}_{\mathcal{S}}$ and $\mathcal{E} \subseteq \mathcal{S}$, where $\mathcal{S}$ is a sample space. If $R_{\cap\mathcal{E}}$ is the restriction operator to $\mathcal{E}$, we have*

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell_{\mathcal{S}}\right] = \frac{\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell_{\mathcal{S}})\right]}{P(\mathcal{E})}\cdot$$

Proof. For an arbitrary $\ell \in \mathcal{A}_{\mathcal{S}}$, let $\ell' = \ell - R_{\cap\mathcal{E}}(\ell)$. Clearly, $R_{\cap\mathcal{E}}(\ell')$ is the zero function, and therefore $\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell')\right] = 0$. Since $\mathcal{A}_{\mathcal{S}}$ is a vector space we have $\ell' \in \mathcal{A}_{\mathcal{S}}$, and the definition of conditional expectation implies $\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell'\right] = 0$. Therefore

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell\right] - \mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[R_{\cap\mathcal{E}}(\ell)\right] = 0\cdot \tag{23}$$

Let $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ represent the range of $R_{\cap\mathcal{E}}$. $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ is a subspace of $\mathcal{A}_{\mathcal{S}}$. By the definition of conditional expectation, functionals obtained by the restrictions of $\mathbb{E}_{\mathcal{S}}$ and $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$ to $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ have the same null space. That implies by Lemma 6 for every $\ell \in \mathcal{A}_{\mathcal{S}}$ we have

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[R_{\cap\mathcal{E}}(\ell)\right] = \lambda\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell)\right]\cdot$$

Thus, by Equation 23

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell\right] = \lambda\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell)\right]\cdot$$

Provided that $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$ is well-defined, for the sure thing $c_{\mathcal{S}}$ we have

$$\lambda = \frac{\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[c_{\mathcal{S}}\right]}{\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(c_{\mathcal{S}})\right]} = \frac{1}{P(\mathcal{E})}\cdot$$

That proves the theorem. □

Definition A.9 (Dual Map)**.** Let $\mathcal{A}$ and $\mathcal{B}$ be two sets. Suppose $M : \mathcal{A} \to \mathcal{B}$ is a map from $\mathcal{A}$ to $\mathcal{B}$. The dual map of M is the map $M^* : \mathbb{R}^{\mathcal{B}} \to \mathbb{R}^{\mathcal{A}}$, defined for each $f \in \mathbb{R}^{\mathcal{B}}$ by

$$M^*(f) = f \circ M\cdot$$

Proposition. The dual map is a linear map.

Theorem 13. *Let $\mathcal{S}$ be a sample space. Suppose $M : \mathcal{S} \to \mathcal{S}'$ is a map from $\mathcal{S}$ to another set $\mathcal{S}'$. If $\mathbb{E}_{\mathcal{S}}$ is an expectation for $\mathcal{S}$, then $\mathbb{E}_{\mathcal{S}'} = M^{**}(\mathbb{E}_{\mathcal{S}})$ is an expectation for $\mathcal{S}'$, where M^{**} is the dual map of M^*, and M^* is the dual map of M.*

Proof. M^{**} is a dual map and therefore is linear.

By definition for every $\ell_{\mathcal{S}'} \in \mathcal{A}_{\mathcal{S}'}$ we have

$$\mathbb{E}_{\mathcal{S}'}\left[\ell_{\mathcal{S}'}\right] = \mathbb{E}_{\mathcal{S}} \circ \ell_{\mathcal{S}'} \circ M\cdot$$

We can easily verify that

$$c_{\mathcal{S}'} \circ M = c_{\mathcal{S}}\cdot$$

Therefore, $\mathbb{E}_{\mathcal{S}'}\left[c_{\mathcal{S}'}\right] = 1$.

On the other hand for any $\mathcal{E}' \subseteq \mathcal{S}'$, we have

$$R_{\cap \mathcal{E}'}(c_{\mathcal{S}'}) \circ M = R_{\cap \mathcal{E}}(c_{\mathcal{S}}),$$

where

$$\mathcal{E} = \{s \in \mathcal{S} : M(s) \in \mathcal{E}'\}\cdot$$

This implies $\mathbb{E}_{\mathcal{S}'}\left[R_{\cap \mathcal{E}'}(c_{\mathcal{S}'})\right] \geq 0$. $\square$

B Latent Regression Model

We consider C classes (or datasets), indexed by $c = 1, \dots, C$. For each class c, we observe a data matrix $X^{(c)} \in \mathbb{R}^{N_c \times M}$, where each row $\boldsymbol{x}_i^{(c)} \in \mathbb{R}^M$ is associated with a D-dimensional latent variable $\boldsymbol{z}_i^{(c)} \in \mathbb{R}^D$.

Thus, each class is associated with a collection of latent variables

$$\boldsymbol{z}^{(c)} = \{\boldsymbol{z}_i^{(c)}\}_{i=1}^{N_c}, \qquad \boldsymbol{z}_i^{(c)} \in \mathbb{R}^D.$$

For each class c, the latent variables are assumed independent and identically distributed according to

$$\boldsymbol{z}_i^{(c)} \sim \mathcal{N}\big(\boldsymbol{\mu}^{(c)}, \tau_c^{-1} I_D\big), \qquad c = 2, \dots, C,$$

where $\boldsymbol{\mu}^{(c)} \in \mathbb{R}^D$ is the class-specific latent mean and $\tau_c > 0$ is a precision parameter controlling the latent dispersion.

To remove redundancy, we fix the latent distribution of the first class as a reference coordinate system. Specifically, we impose

$$\boldsymbol{\mu}^{(1)} = \mathbf{0}, \qquad \tau_1 = 1,$$

so that

$$\boldsymbol{z}_i^{(1)} \sim \mathcal{N}(\mathbf{0}, I_D).$$

This anchoring removes global translation and scaling symmetries without restricting the expressive capacity of the model. The latent geometry of each class is therefore interpreted relative to the reference class $c = 1$.

Let $\phi : \mathbb{R}^D \to \mathbb{R}^K$ denote a fixed nonlinear basis mapping. The design matrix of class c is defined as $Z^{(c)} \in \mathbb{R}^{N_c \times K}$ with entries

$$Z_{ik}^{(c)} = \phi_k(\boldsymbol{z}_i^{(c)}).$$

We collect the decoder weights in a shared matrix

$$W = [\boldsymbol{w}_1 \ \dots \ \boldsymbol{w}_M] \in \mathbb{R}^{K \times M},$$

where each column $\boldsymbol{w}_j \in \mathbb{R}^K$ corresponds to the weights for the j-th observed dimension across all classes.

Conditioned on $\boldsymbol{z}^{(c)}$ and W, observations are generated as

$$p(X^{(c)} \mid \boldsymbol{z}^{(c)}, W) = \prod_{i=1}^{N_c} \prod_{j=1}^{M} \mathcal{N}\left(x_{ij}^{(c)} \,\middle|\, \phi(\boldsymbol{z}_i^{(c)})^\top \boldsymbol{w}_j, \beta_j^{-1}\right),$$

or equivalently,

$$\boldsymbol{x}_j^{(c)} \sim \mathcal{N}\big(Z^{(c)}\boldsymbol{w}_j, \beta_j^{-1} I_{N_c}\big)\,,$$

where $\boldsymbol{x}_j^{(c)}$ denotes the j-th column of $X^{(c)}$ and $\beta_j > 0$ is the observation precision of that column.

To regularize the decoder, we assume noninformative priors for the latent parameters $\{\boldsymbol{\mu}^{(c)}, \tau_c\}_{c=2}^{C}$, and place independent Gaussian priors on the columns of W:

$$p(W) = \prod_{j=1}^{M} \mathcal{N}\left(\boldsymbol{w}_j \;\middle|\; \boldsymbol{0}, \alpha_j^{-1} I_K\right).$$

The full joint distribution factorizes as

$$p(W)\prod_{c=1}^{C} p(X^{(c)} \mid \boldsymbol{z}^{(c)}, W)p(\boldsymbol{z}^{(c)} \mid \boldsymbol{\mu}^{(c)}, \tau_c)p(\boldsymbol{\mu}^{(c)}, \tau_c).$$

This defines a shared nonlinear generative model in which all classes are coupled through the common decoder W, while retaining class-specific latent geometry via $(\boldsymbol{\mu}^{(c)}, \tau_c)$.

B.1 Variational Inference for the Single-Class Case

To simplify the exposition, we now consider the single-class case. Since only one class is present, we remove the superscript (c) from the notation for clarity.

Furthermore, we restrict the latent variable to be scalar,

$$\boldsymbol{z} = \{z_i\}_{i=1}^{N} \qquad z_i \in \mathbb{R},$$

so that

$$p(\boldsymbol{z} \mid \mu, \tau) = \prod_{i=1}^{N} \mathcal{N}(z_i \mid \mu, \tau^{-1}).$$

We treat μ and τ as unknown parameters to be inferred, using the standard joint Jeffreys prior for the mean and precision of a Gaussian model,

$$p(\mu, \tau) \propto \tau^{-1}.$$

The nonlinear basis mapping reduces to

$$\phi : \mathbb{R} \to \mathbb{R}^K,$$

and the design matrix $Z \in \mathbb{R}^{N \times K}$ is defined by

$$Z_{ik} = \phi_k(z_i).$$

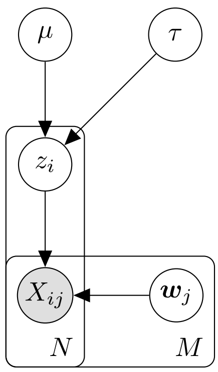

Figure 3: One class scalar latent regression model

We see the Bayesian network of this model in Figure 3. Exact posterior inference in this model is intractable due to the nonlinear dependence of the likelihood on $\boldsymbol{z}$. We therefore proceed by introducing a variational approximation.

In order to formulate a variational treatment of this model, we consider a variational distribution which factorizes between the latent variables and the parameters so that

$$q(\boldsymbol{z}, W, \mu, \tau) = q(\boldsymbol{z})q(W, \mu, \tau).$$

First, let us consider the derivation of the update equation for the factor $q(W, \mu, \tau)$. The log of the optimized factor is given by

$$\ln q^*(W, \mu, \tau) = \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(W, \mu, \tau \mid \boldsymbol{z}, X)\right] + \text{const.}$$

As can be seen from Figure 3, the d-separation criterion implies that

$$\mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(W, \mu, \tau \mid \boldsymbol{z}, X)\right] = \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\mu, \tau \mid \boldsymbol{z})\right] + \sum_j \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\boldsymbol{w}_j \mid \boldsymbol{z}, X)\right].$$

Therefore, we have the following induced factorization for $q(W, \mu, \tau)$:

$$q(W, \mu, \tau) = q(\mu, \tau)\prod_{j=1}^{M} q(\boldsymbol{w}_j).$$

We obtain $q(\boldsymbol{w}_j)$ using the fact that each $\boldsymbol{w}_j$ is a weight vector of a Bayesian linear regression model:

$$\begin{aligned}\ln p(\boldsymbol{w}_j \mid \boldsymbol{z}, X) &= -\frac{\beta_j}{2}\left\|Z\boldsymbol{w}_j - \boldsymbol{x}_j\right\|^2 - \frac{\alpha_j}{2}\boldsymbol{w}_j^\top \boldsymbol{w}_j + \text{const} \\ &= -\frac{1}{2}\boldsymbol{w}_j^\top(\beta_j Z^\top Z + \alpha_j I)\boldsymbol{w}_j + \beta_j \boldsymbol{x}_j^\top Z\boldsymbol{w}_j + \text{const.}\end{aligned}$$

Taking the expectation with respect to $q(\boldsymbol{z})$, we obtain

$$\ln q^*(\boldsymbol{w}_j) = -\frac{1}{2}\boldsymbol{w}_j^\top(\beta_j \mathbb{E}\left[Z^\top Z\right] + \alpha_j I)\boldsymbol{w}_j + \beta_j \boldsymbol{w}_j^\top \mathbb{E}\left[Z\right]^\top \boldsymbol{x}_j + \text{const},$$

where expectations of matrices are taken element-wise. The resulting expression is quadratic in $\boldsymbol{w}_j$. Completing the square yields

$$q^*(\boldsymbol{w}_j) = \mathcal{N}(\boldsymbol{w}_j \mid \boldsymbol{\mu}_j, \Sigma_j).$$

The corresponding moments are

$$\begin{aligned} \mathbb{E}\left[\boldsymbol{w}_j\right] &= \boldsymbol{\mu}_j, \\ \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top\right] &= \boldsymbol{\mu}_j \boldsymbol{\mu}_j^\top + \Sigma_j, \end{aligned}$$

with

$$\begin{aligned} \Sigma_j^{-1} &= \beta_j \mathbb{E}\left[Z^\top Z\right] + \alpha_j I, \\ \boldsymbol{\mu}_j &= \beta_j \Sigma_j \mathbb{E}\left[Z\right]^\top \boldsymbol{x}_j. \end{aligned}$$

Since the design matrix satisfies $Z_{ik} = \phi_k(z_i)$, we have

$$\mathbb{E}\left[Z^\top Z\right] = \sum_{i=1}^{N} \mathbb{E}\left[\phi(z_i)\phi(z_i)^\top\right].$$

Collecting the posterior means into a matrix gives

$$\mathbb{E}\left[W\right] = [\boldsymbol{\mu}_1 \ \ldots \ \boldsymbol{\mu}_M].$$

Moreover, to simplify later expressions, define the diagonal precision matrix

$$B = \operatorname{diag}(\beta_1, \ldots, \beta_M).$$

Then, we have

$$\sum_{j=1}^{M} \beta_j \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top\right] = \mathbb{E}\left[W\right] B \mathbb{E}\left[W\right]^\top + \sum_{j=1}^{M} \beta_j \Sigma_j.$$

This quantity will appear in the update for the latent variables. Note that if $\alpha_j = 0$ for all j, the mean simplifies to

$$\mathbb{E}\left[W\right] = \mathbb{E}\left[Z^\top Z\right]^{-1} \mathbb{E}\left[Z\right]^\top X.$$

We now consider the update equation for the factor $q(\mu, \tau)$:

$$\begin{aligned}
\ln q^*(\mu, \tau) &= \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\mu, \tau \mid \boldsymbol{z})\right] + \text{const} \\
&= -\frac{\tau}{2}\left(\sum_i \mathbb{E}\left[z_i^2\right] - 2\mu \sum_i \mathbb{E}\left[z_i\right] + N\mu^2\right) \\
&\quad + \left(\frac{N}{2} - 1\right)\ln \tau + \text{const.}
\end{aligned}$$

We define

$$\begin{aligned}
\overline{z} &= \frac{1}{N}\sum_i \mathbb{E}\left[z_i\right], \\
\overline{zz} &= \frac{1}{N}\sum_i \mathbb{E}\left[z_i^2\right].
\end{aligned}$$

We observe that $q^*(\mu, \tau)$, for $N > 1$ follows a Normal–Inverse–Gamma distribution:

$$q^*(\mu, \tau) = \mathcal{N}\left(\mu \,\middle|\, \overline{z}, \frac{1}{N\tau}\right) \text{Gamma}\left(\tau \,\middle|\, \frac{N-1}{2}, \frac{N}{2}(\overline{zz} - \overline{z}^2)\right).$$

This implies

$$\begin{aligned}
\mathbb{E}\left[\tau\right] &= \frac{(N-1)}{N(\overline{zz} - \overline{z}^2)}, \\
\mathbb{E}\left[\mu\tau\right] &= \overline{z}\mathbb{E}\left[\tau\right].
\end{aligned}$$

Finally, we derive the update equation for $q(\boldsymbol{z})$:

$$\begin{aligned}
\ln q^*(\boldsymbol{z}) &= \mathbb{E}_{q(W, \mu, \tau)}\left[\ln p(X \mid \boldsymbol{z}, W) + \ln p(\boldsymbol{z} \mid \mu, \tau)\right] + \text{const} \\
&= \mathbb{E}_{q(W)}\left[\ln p(X \mid \boldsymbol{z}, W)\right] + \mathbb{E}_{q(\mu, \tau)}\left[\ln p(\boldsymbol{z} \mid \mu, \tau)\right] + \text{const.}
\end{aligned}$$

We expand the terms $\ln p(X \mid \boldsymbol{z}, W)$ and $\ln p(\boldsymbol{z} \mid \mu, \tau)$. All expressions that are independent of $\boldsymbol{z}$ are treated as constants.

$$
\begin{aligned}
\ln p(X \mid \boldsymbol{z}, W) &= \sum_j \ln \mathcal{N}(\boldsymbol{x}_j \mid Z\boldsymbol{w}_j, \beta_j^{-1} I) \\
&= -\frac{1}{2} \sum_j \beta_j \left\| Z\boldsymbol{w}_j - \boldsymbol{x}_j \right\|^2 + \text{const} \\
&= \sum_i \left(-\frac{1}{2} \phi(z_i)^\top \sum_j \beta_j \boldsymbol{w}_j \boldsymbol{w}_j^\top \phi(z_i) + \phi(z_i)^\top \sum_j \beta_j X_{ij} \boldsymbol{w}_j \right) + \text{const.}
\end{aligned}
$$

$$
\begin{aligned}
\ln p(\boldsymbol{z} \mid \mu, \tau) &= -\frac{\tau}{2} \sum_i (z_i - \mu)^2 + \text{const} \\
&= \sum_i \left(-\frac{\tau}{2} z_i^2 + \tau \mu z_i \right) + \text{const.}
\end{aligned}
$$

Substituting these expansions into the variational update expression and taking the expectation yields:

$$
\ln q^*(\boldsymbol{z}) = \sum_i \ln q^*(z_i),
$$

where

$$
\begin{aligned}
\ln q^*(z_i) = &\left(-\frac{1}{2} \phi(z_i)^\top \sum_j \beta_j \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top \right] \phi(z_i) + \phi(z_i)^\top \mathbb{E}\left[W \right] B X_{i,:}^\top \right) \\
&+ \left(-\frac{1}{2} \mathbb{E}\left[\tau \right] z_i^2 + \mathbb{E}\left[\tau \mu \right] z_i \right) + \text{const.}
\end{aligned}
$$

Here $X_{i,:}$ denotes the i-th row of the observation matrix X.

Because the update expression separates as a sum over i, the optimal factor $q(\boldsymbol{z})$ factorizes as $\prod_i q(z_i)$. Define

$$
\begin{aligned}
S^{-1} &= \sum_j \beta_j \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top \right], \\
\boldsymbol{m}_i &= S \mathbb{E}\left[W \right] B X_{i,:}^\top .
\end{aligned}
$$

Then, up to normalization,

$$
q^*(z_i) \propto \mathcal{N}\left(\phi(z_i) \mid \boldsymbol{m}_i, S \right) \mathcal{N}\left(z_i \;\middle|\; \mathbb{E}\left[\mu \right], \mathbb{E}\left[\tau \right]^{-1} \right).
$$

Note that since the basis mapping is nonlinear, this does not generally imply that $q^*(z_i)$ is Gaussian in z_i.

As seen previously, the update equations require expectations of the form

$$\mathbb{E}\left[\phi_i(z_k)\phi_j(z_k)\right], \quad \mathbb{E}\left[\phi_i(z_k)\right], \quad \mathbb{E}\left[z_k\right], \quad \mathbb{E}\left[z_k^2\right],$$

for all relevant indices i, j, k. These expectations typically do not admit closed forms and must be approximated numerically.